

Identifikasi Support dan Resistance pada Saham Sektor Energi IDX Menggunakan Metode K-Means Clustering dan Backtesting

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Abstrak— Sektor energi di Bursa Efek Indonesia (BEI) memiliki tingkat volatilitas yang tinggi karena sangat dipengaruhi oleh gejolak harga komoditas global dan ketidakpastian kebijakan domestik. Analisis teknikal konvensional dalam menentukan level *support* dan *resistance* (S/R) sering kali dikritik karena sifatnya yang subjektif dan inkonsistensi antar analis. Penelitian ini bertujuan mengoptimalkan penentuan batas zona *support* dan *resistance* (S/R) saham sektor energi menggunakan pendekatan K-Means Clustering dan formulasi rentang dinamis (*dynamic width*). Pengujian performa dilakukan secara komparatif menggunakan skenario nilai K=2 hingga K=10 melalui metode *algorithmic out-of-sample backtesting* terhadap data historis harian periode tahun 2019 sampai 2025 pada 26 emiten sektor energi yang memenuhi kriteria *intersection baseline*. Hasil eksperimen empiris menunjukkan adanya fenomena *trade-off* spasial yang konsisten, di mana peningkatan nilai K secara eksponensial mempersempit ketebalan zona sehingga menurunkan nilai *Recall* secara drastis hingga menyentuh angka 18,88% pada skenario K=10. Berdasarkan metrik *Confusion Matrix*, Model K=2 dinobatkan sebagai konfigurasi spasial paling optimal dengan pencapaian puncak rata-rata *F1-Score* sektoral sebesar 67,28%, tingkat *Precision* 86,25%, *Recall* 57,34%, dan *Accuracy* 73,11%. Temuan ini membuktikan bahwa pembagian area psikologis bursa secara makro dengan batas berbasis deviasi internal jauh lebih andal dan adaptif dalam menangkap peluang pembalikan (*reversal*) harga saham sektor energi.

Kata kunci— K-Means Clustering, Support Resistance, Backtesting, IDX.

I. PENDAHULUAN

Sektor energi di Bursa Efek Indonesia (BEI) yang didominasi komoditas minyak, gas, dan batu bara memiliki karakteristik pergerakan harga yang sangat fluktuatif. Pola pergerakan harga pada sektor ini sangat sensitif terhadap perubahan harga komoditas dunia, regulasi pemerintah, serta sentimen pasar global (Awan et al., 2023). Selain faktor ekonomi makro, gejolak pada pasar keuangan di negara berkembang sering kali diperparah oleh dinamika politik domestik dan ketidakpastian arah kebijakan negara (Ahir et al., 2020). Fenomena tersebut terlihat jelas pada pergerakan Indeks Harga Saham Gabungan (IHSG) saat merespons pengumuman pasangan calon presiden dan wakil presiden Indonesia yang memicu reaksi pelemahan akibat meningkatnya sikap kehati-hatian investor (Rochimah & Yuliana, 2024). Ditambah dengan bergulirnya proses transisi energi nasional, ketidakpastian terhadap kinerja masa depan emiten energi semakin memperkuat risiko investasi (Sisdwinugraha et al., 2025). Tingginya risiko fluktuasi harga ini menuntut penggunaan metode analisis yang presisi, di

mana salah satu acuan utamanya adalah penentuan area pembalikan arah pergerakan saham.

Dalam disiplin analisis teknikal, penentuan titik balik harga tersebut bertumpu pada dua konsep dasar, yaitu *support* dan *resistance*. *Support* didefinisikan sebagai area harga di mana akumulasi tekanan beli cukup kuat menahan penurunan, sedangkan *resistance* merupakan area di mana tekanan jual membatasi tren kenaikan harga (I Kadek Dwik Darmawan & Henny Rahyuda, 2024). Kedua level ini menjadi acuan krusial bagi investor untuk menentukan momentum masuk (*entry*), keluar (*exit*), hingga membatasi risiko kerugian melalui *stop loss* (Wealthsimple, 2025). Kendati perannya sangat vital, mayoritas praktisi masih menentukan level S/R secara konvensional, yakni menarik garis lurus manual pada titik tertinggi atau terendah visual (Saqan, 2021). Pendekatan tradisional ini dinilai sangat subjektif karena analis yang berbeda dapat menghasilkan interpretasi level harga yang berbeda pada grafik saham yang sama (Sim, 2020). Kelemahan aspek subjektivitas pada penarikan garis manual ini kemudian mendorong urgensi pemanfaatan komputasi cerdas guna menghasilkan kalkulasi yang lebih objektif.

Guna mengatasi keterbatasan tersebut, pendekatan berbasis *machine learning* mulai dikembangkan melalui penerapan algoritma *K-Means clustering* untuk mengelompokkan data harga historis secara otomatis (Öztürk, 2024). Performa algoritma berbasis jarak seperti *K-Means* ini terbukti meningkat ketepatannya secara signifikan setelah melalui proses standarisasi data menggunakan *Z-score* (Wongoutong, 2024). Namun, kelemahan utama dari *K-Means* standar adalah *output*-nya yang hanya berupa titik koordinat tunggal, padahal pada realitasnya harga saham selalu memantul di dalam suatu rentang area, bukan angka mati. Oleh karena itu, penelitian ini mengusulkan sebuah *Hybrid Machine Learning Pipeline* melalui pengembangan *Dynamic Spatial Boundary Model* yang memformulasikan ketebalan zona memanfaatkan nilai standar deviasi internal klaster ($\sigma_{internal}$) demi menyaring kebisingan pasar (*market noise*). Melalui pengujian parameter K=2 hingga K=10, efektivitas model ini dievaluasi menggunakan simulasi *backtesting* algoritmik guna mengunci konfigurasi model spasial terbaik berdasarkan metrik *Confusion Matrix*.

II. METODE

Use either SI (MKS) or CGS as primary units. (SI units are strongly encouraged.) The SI unit for magnetic field strength

H is A/m.)” An exception is when English units are used as identifiers in trade, such as “14 in laptop.” Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity in an equation.

III. FIGURES AND TABLES

Large figures and tables may span both columns. Place figure captions below the figures; place table titles above the tables. Please do not include captions as part of the figures. Do not put captions in "text boxes" linked to the figures. Do not put borders around the outside of your figures.

For paper in English, use the abbreviation "Fig." even at the beginning of a sentence. Do not abbreviate "Table.". If paper is written in Bahasa Indonesia, use "Gambar. " and "Tabel." as usual.

Fig. 1. SRM geometry with FEM mesh: 6/6 stator/rotor

Photographs and greyscale figures should be prepared with 220 dpi resolution and saved with no compression, 8 bits per pixel (greyscale). Colour figures should be prepared with 400 dpi resolution and saved with no compression, 8 bits per pixel (palette or 256 colour) minimum.

Fig. 2. Torque characteristics versus rotor position and different current density of exciting current: 6/6 stator/rotor

Tables and figures should be centered. Place table captions above the tables (Table I) and figure captions below the figures (Figures 1 and 2). Use *Table Title Style* for table caption and use *Figure Captions Style* for figure. Figures and tables may

span across both columns (Fig.1). Tables are numbered with Roman numerals (*Table Title style*, 8 pt, centered).

IV. MATH AND UNITS

A. Math

Number equations consecutively with equation numbers in parentheses flush with the right margin, as in (1). If you are using *Word*, use either the Microsoft Equation Editor or the *MathType* add-on (<http://www.mathtype.com>) for equations in your paper (Insert | Object | Create New | Microsoft Equation or MathType Equation). “Float over text” should *not* be selected.

Expressions and equations must be centred. They need to be numbered as follows:

$$U = R I(1)$$

References within the text will have the number of the equation within curved parenthesis, as in the case of (1). Variables will be in *Italics*, constants in *Normal*, and matrices in **Bold**.

$$U = Z \cdot I \quad (2)$$

$$\cos \cos \alpha + \cos \cos \beta = 2 \cos \cos \frac{1}{2}(\alpha + \beta)$$

$$\cos \cos \frac{1}{2}(\alpha - \beta) \quad (3)$$

B. Units

Use either SI (MKS) or CGS as primary units. (SI units are strongly encouraged.) English units may be used as secondary units (in parentheses). **This applies to papers in data storage.** For example, write “15 Gb/cm² (100 Gb/in²).” An exception is when English units are used as identifiers in trade, such as “3½ in disk drive.” Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity in an equation.

The SI unit for magnetic field strength H is A/m. However, if you wish to use units of T, either refer to magnetic flux density B or magnetic field strength symbolized as $\mu_0 H$. Use the center dot to separate compound units, e.g., “A·m².”

V. REFERENCE AND CITATION

Number citations consecutively in square brackets [1]. The sentence punctuation follows the brackets [2]. Multiple references [2], [3] are each numbered with separate brackets [1]-[3]. When citing a section in a book, please give the relevant page numbers [2]. In sentences, refer simply to the reference number, as in [3]. Do not use "Ref. [3]" or "reference [3]" except at the beginning of a sentence: "Reference [3] shows"

The Reference list must appear numbered in square brackets

in the last (non numbered) section and should have the following structure:

- Author(s)
- Title of the work or article
- Journal, conference, book or thesis, in italics
- Publisher, in the case of books
- Page numbers if necessary (example: pp.32-36)
- Place and date of publication (with books and theses, the year of publication is sufficient).

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

This statement is **mandatory** and must be included in the manuscript. Authors are required to follow the wording provided below:

The authors declare that generative artificial intelligence tools were used solely for language editing and grammatical improvement. The authors take full responsibility for the content, interpretation, and conclusions presented in this manuscript.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in American English is without an “e” after the “g.” Use the singular heading even if you have many acknowledgments. Avoid expressions such as “One of us (S.B.A.) would like to thank” Instead, write “F. A. Author thanks”

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TABLE I

UNITS FOR MAGNETIC PROPERTIES

No vertical lines in table. Statements that serve as captions for the entire table do not need footnote letters.

^aGaussian units are the same as cgs emu for magnetostatics; Mx = maxwell, G = gauss, Oe = oersted; Wb = weber, V = volt, s = second, T = tesla, m = meter, A = ampere, J = joule, kg = kilogram, H = henry.